

GT2N: Open-Source 2nm Nanosheet PDK and SRAM Memory Compiler

Dongwon Jang

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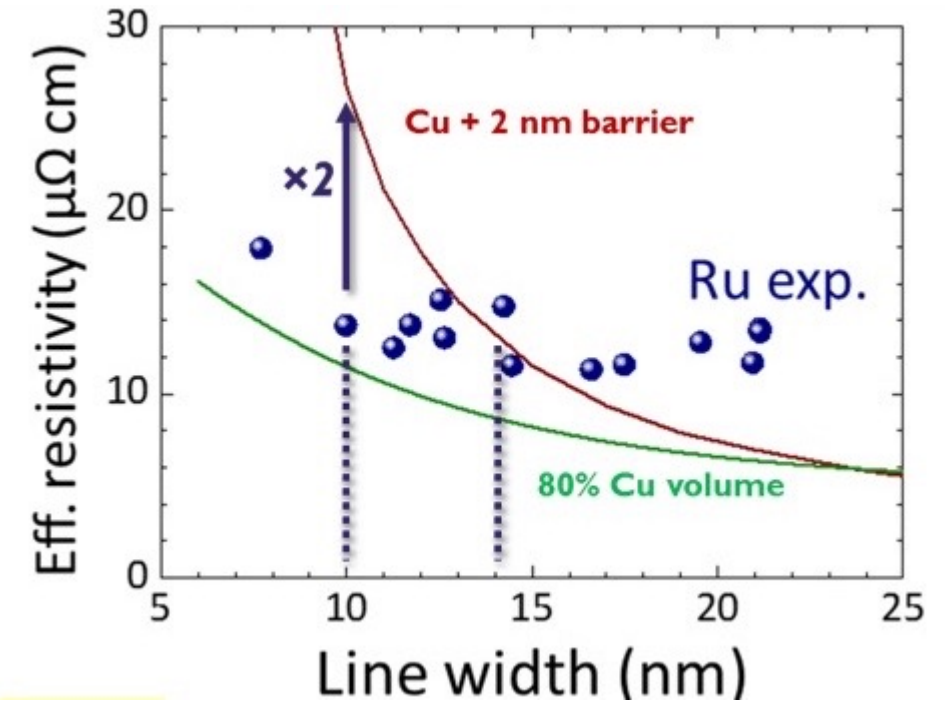
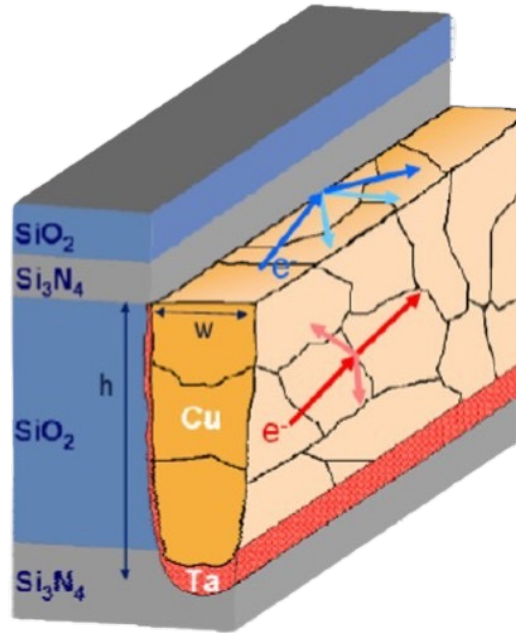
Date: Jul. 28th 2026

2nm PDK Design

Copper Can Scale No Further in 2nm Nodes

Cu interconnects below 2nm nodes:

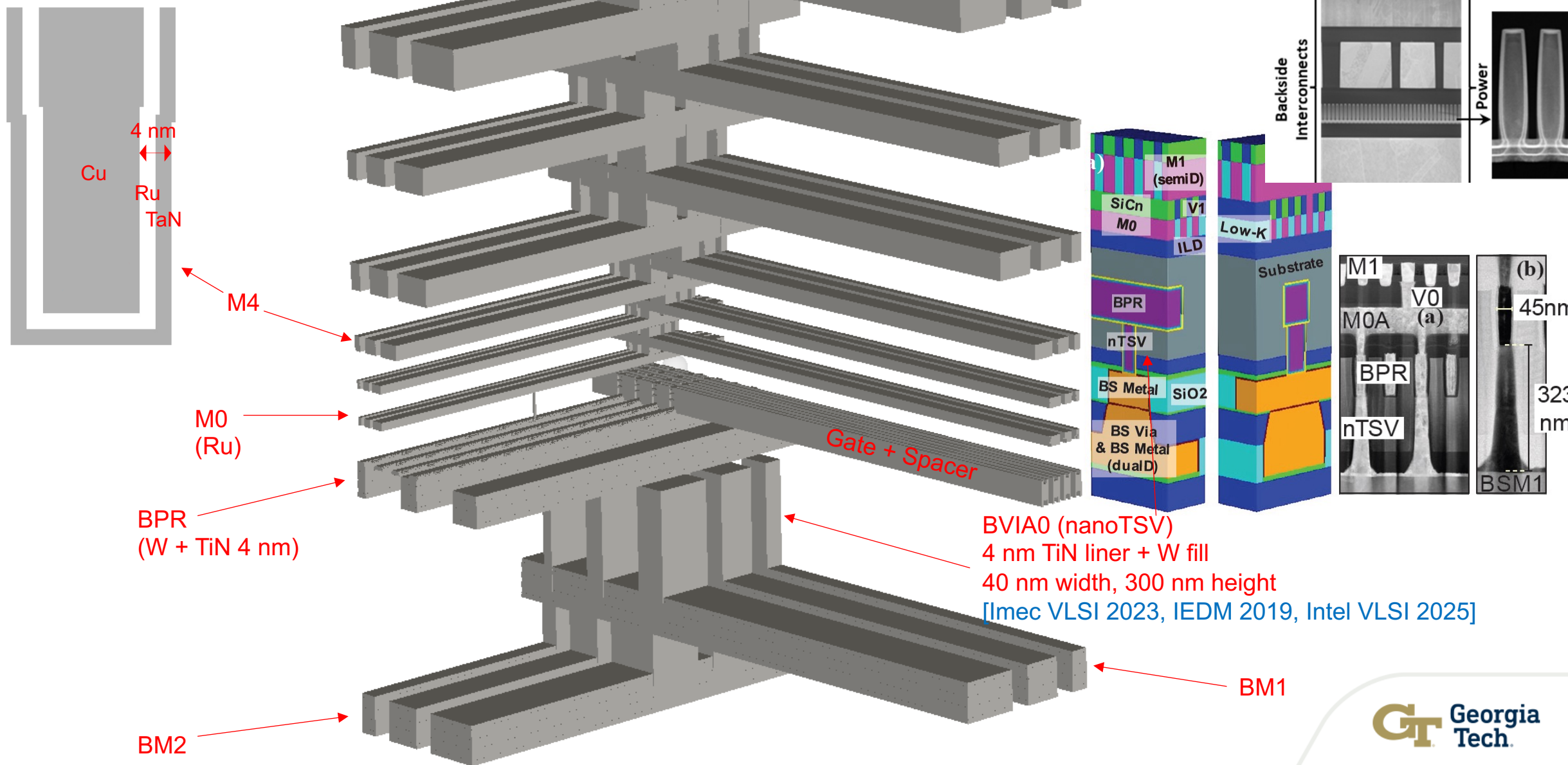
- Electron surface scattering
- Grain boundary effects
- Thick barrier layers reducing conductive volume
- Increasing electromigration risks



→ **Barrierless metal (Ru)** is the future for 2nm and beyond

[Imec Technology Forum 2017]

Interconnect 3D Modeling



Interconnect 3D Modeling

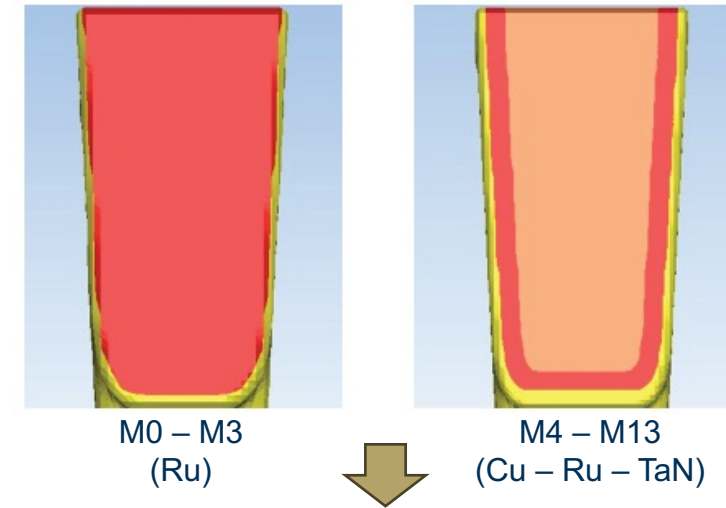
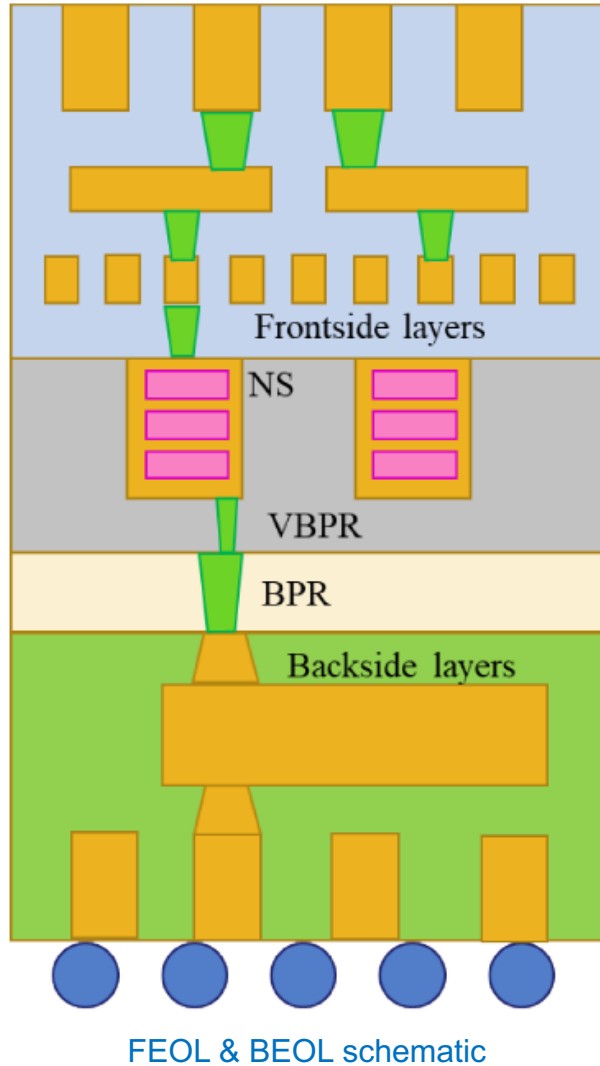


TABLE I

ASSUMPTIONS ON BEOL LITHOGRAPHY AND KEY DIMENSIONS FOR EACH LAYER.

Layer	Width/Pitch	Resistance	Lithography
M0/M2	12/24	622 $\Omega/\mu m$	EUV SPT 0.55NA
V0/V2	12/24	55 Ω	EUV SPT 0.55NA
M1/M3	14/28	438 $\Omega/\mu m$	EUV SPT 0.55NA
V1/V3	14/28	46 Ω	EUV SPT 0.55NA
M4-M5	21/42	166 $\Omega/\mu m$	SADP
V4-V5	21/42	41 Ω	SADP
M6-M9	38/76	26 $\Omega/\mu m$	193i SPT
V6-V9	38/76	13 Ω	193i
M10-M11	56/112	7.5 $\Omega/\mu m$	193i SPT
V10-V11	56/112	6 Ω	193i
M12-M13	360/720	0.64 $\Omega/\mu m$	193i SPT
V12-V13	56/112	6 Ω	193i

Design Rules

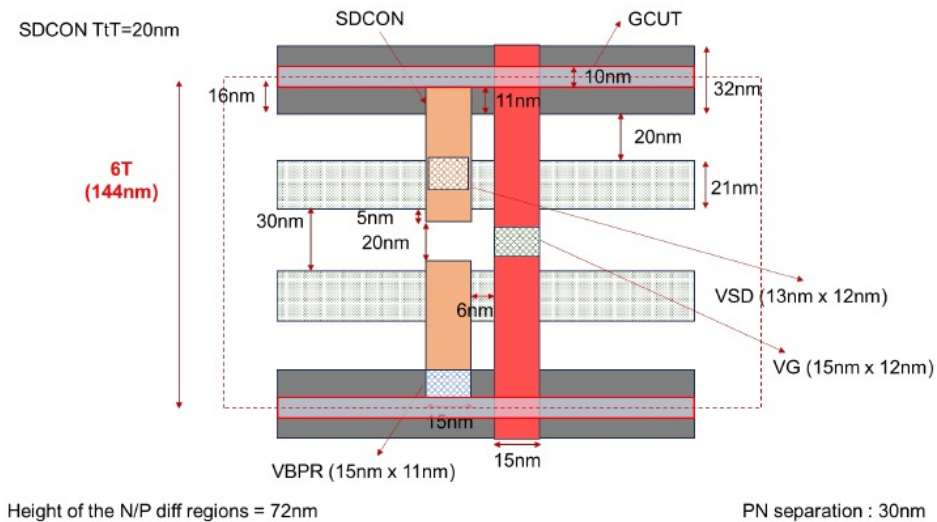


Fig. 4. Standard cell design template with key FEOL/MOL design rules.

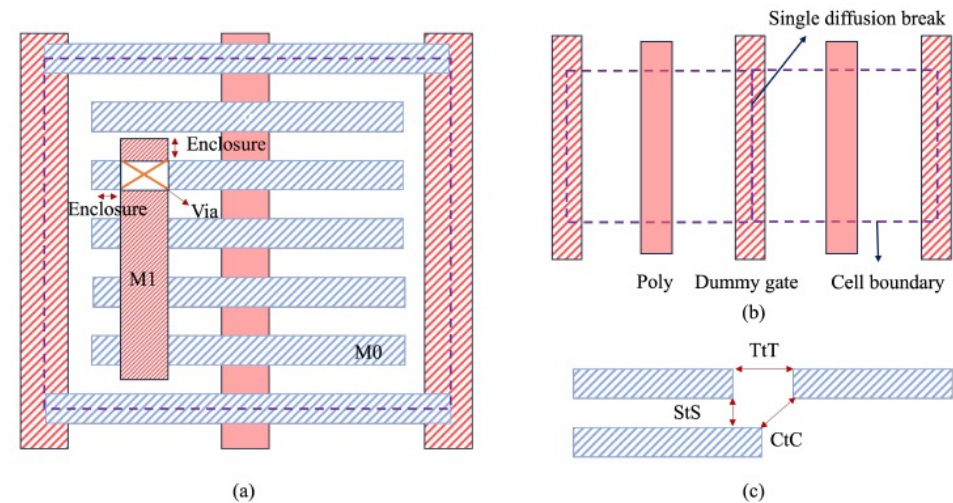
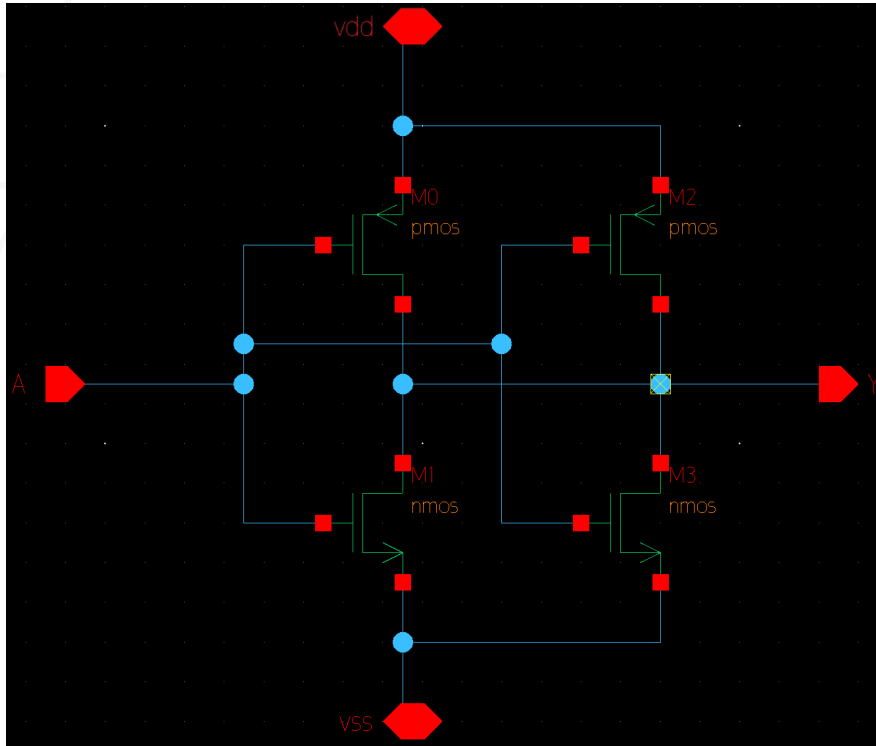


Fig. 5. (a) M0 track locations inside the cells along with the description for via enclosure rules. (b) Illustration of SDB isolation of neighboring cells. (c) Description of various BEOL spacing design rules.

Standard Cell Library Schematic Design



INVx2 cell schematic

LVS Compare Results: PASS

DRC and Extraction Results: CLEAN

[gt3_6t_inv_x2, gt3_6t_inv_x2]

Netlist Extraction Statistics

Library name: /usr/scratch/djang62/GT3_3nm_PDK/synopsys_custom/gt3_6t_inv_x2.icv.lvs/gt3_6t_inv_x2.custom_com
Structure name: gt3_6t_inv_x2
Generated by: IC Validator RHEL64 V-2023.12-SP2.9906768 2024/03/01
Runset name: /usr/scratch/djang62/GT3_3nm_PDK/icv_runset/gt3_main.lvs.rs
User name: djang62
Time started: 2024/08/02 12:49:17AM
Time ended: 2024/08/02 12:49:21AM

Called as: icv -f gdsii -i /usr/scratch/djang62/GT3_3nm_PDK/synopsys_custom/gt3_6t_inv_x2.icv.lvs/gt3_6t_inv_x2.

Layout vs. Schematic Statistics

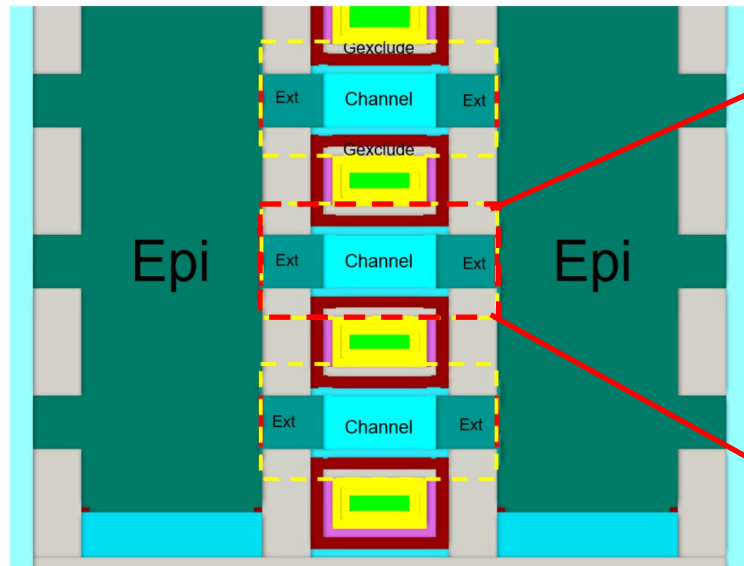
Schematic: /usr/scratch/djang62/GT3_3nm_PDK/synopsys_custom/gt3_6t_inv_x2.icv.lvs/gt3_6t_inv_x2.sch_out.gz

LVS Errors:

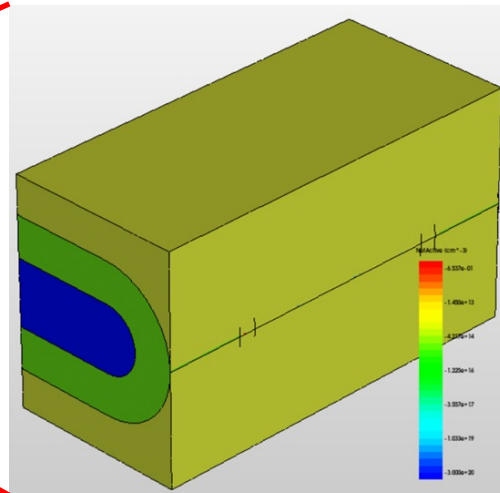
blackbox cells checked: 0
passed 0
failed 0
equivalence points checked: 1
passed 1
failed 0

INVx2 cell layout DRC, LVS validation

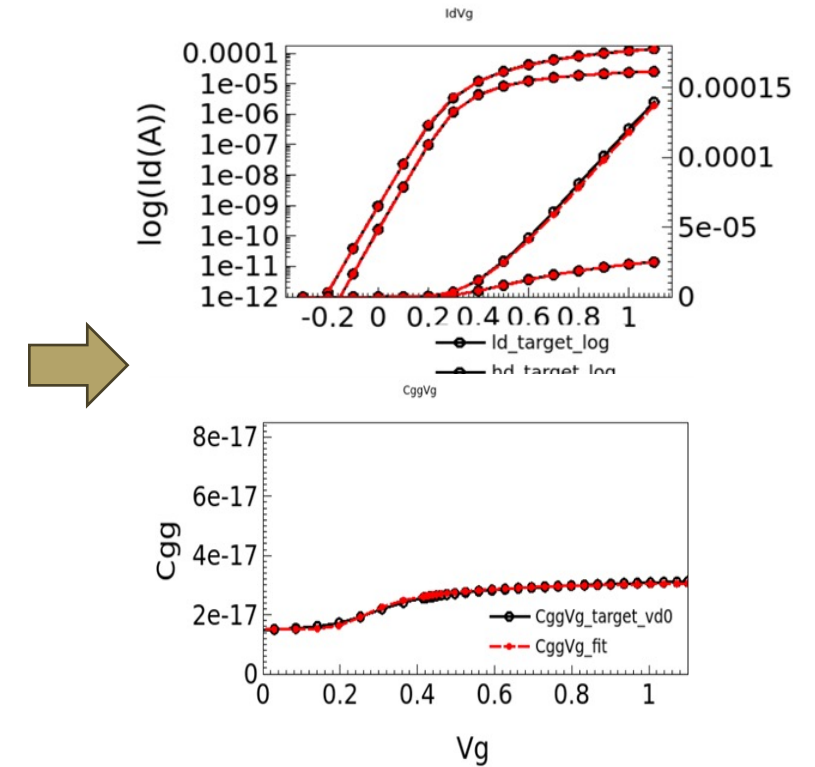
Nanosheet Device 3D Modeling



3-sheet device schematic



Single nanosheet 3D modeling



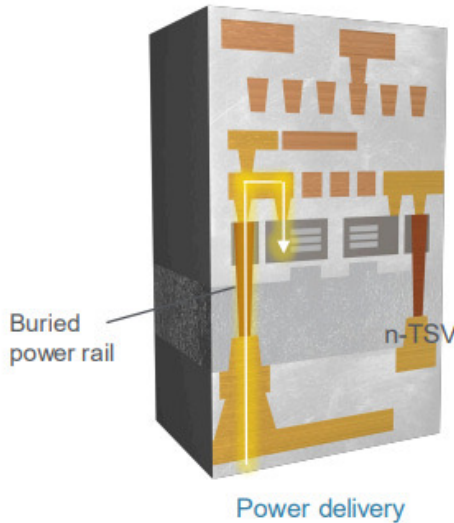
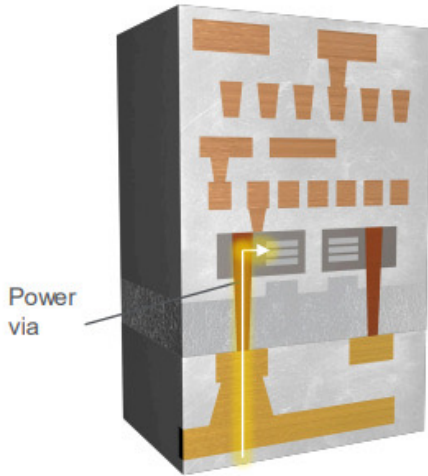
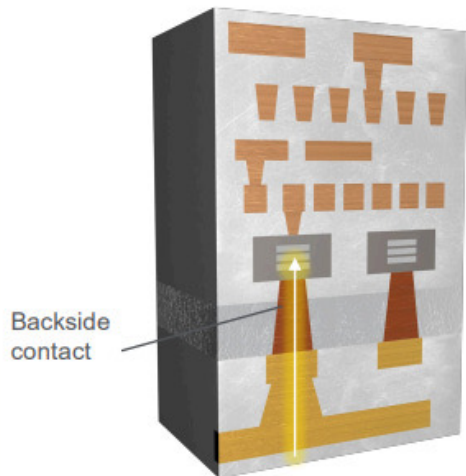
Single nanosheet SPICE modeling

The device model accounts for the RC characteristics of only a single sheet, instead of the entire device region.

Process-Aware Modeling Framework

Power Delivery Architectures

Announced Backside Power Distribution Network Approaches

	Buried Power Rail	Power Via	Backside Contact to S/D
			
Area Scaling	Good	Better	Best
Process Complexity	Low	Medium	High
	Source: Adapted from Song et al (IEDM, 2021) with subsequent analysis by Applied Materials	Source: Adapted from public company disclosures (2021) with subsequent analysis by Applied Materials	Source: Adapted from Song et al (IEDM, 2021) with subsequent analysis by Applied Materials

GT2 Power Delivery Architecture

Contact resistivity

Silicide-nSi: $2.2\text{e-}9 \Omega \text{ cm}^2$

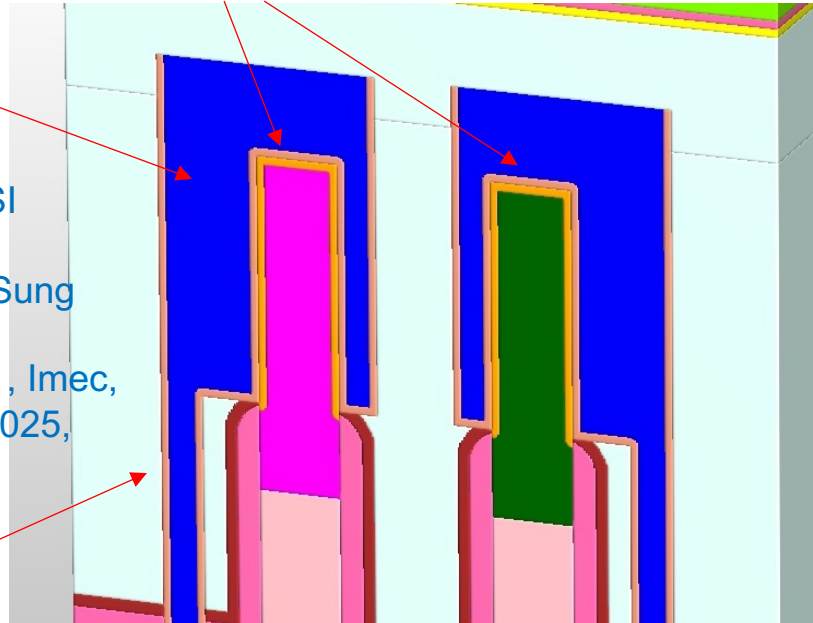
Silicide-pSiGe: $2.0\text{e-}9 \Omega \text{ cm}^2$

[IBM & GF, IEDM 2017, IEDM 2016, TNANO 2019,
Applied Materials, VLSI 2017]

Wrap-around contact
(2 nm TiN liner + W fill)

[Applied Materials, VLSI
2017, SISPAD 2023,
IBM VLSI 2017, W.-L. Sung
et al. TED 2021]

[Qualcomm IEDM 2021, Imec,
VLSI 2025, ESSERC 2025,
Postech VLSI 2025]



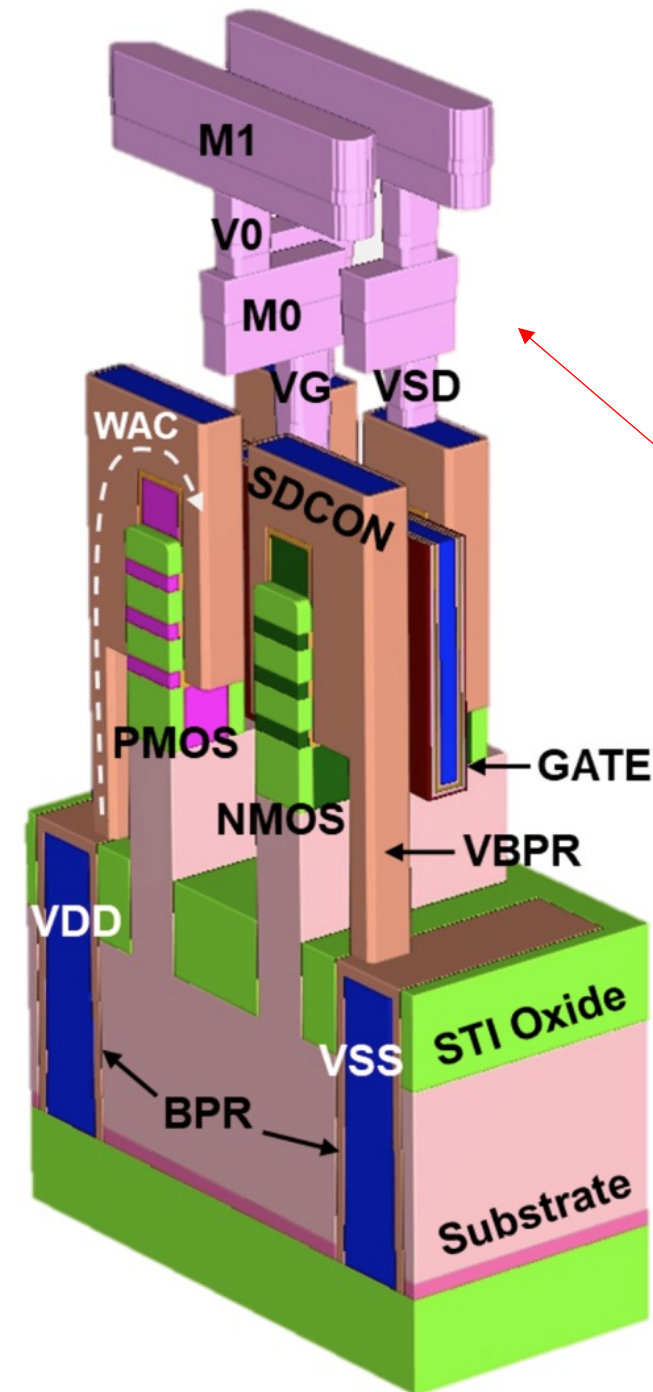
VBPR nanoTSV
(2 nm TiN liner + W fill)

[Imec, IITC 2018, EDTM 2019]

BPR

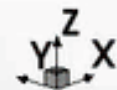
(4 nm TiN liner + W fill)

[Imec, IEDM 2021, VLSI 2022, VLSI 2023,
SPIE DTCO 2023, ESSERC 2025]



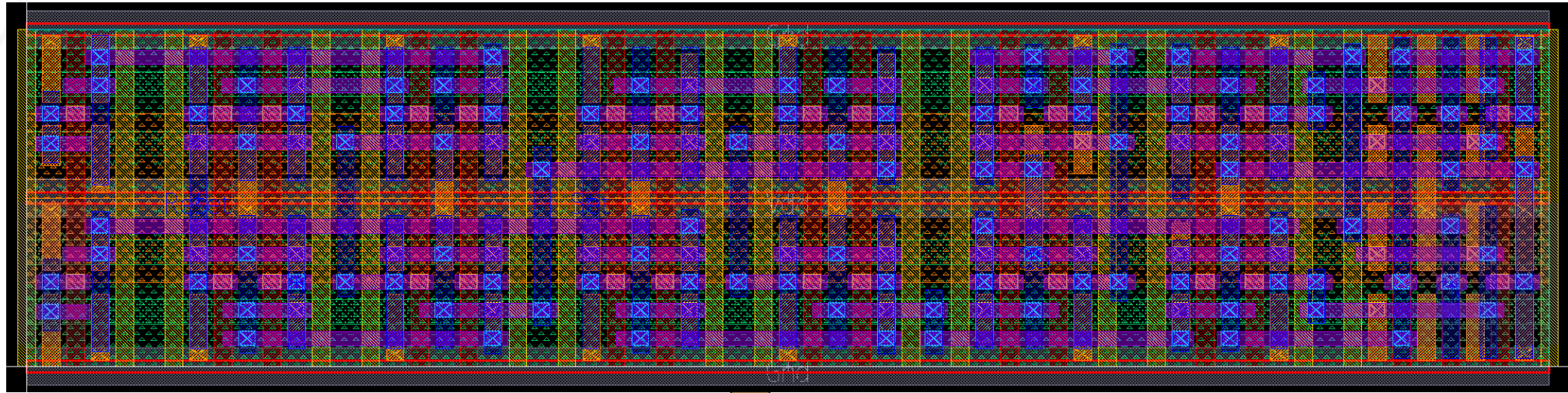
Ru MOL, BEOL

View

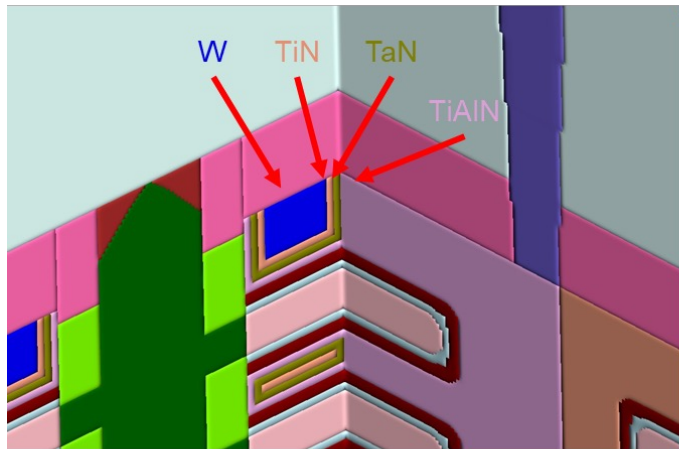


GitHub/azadnaeemi/GT2N

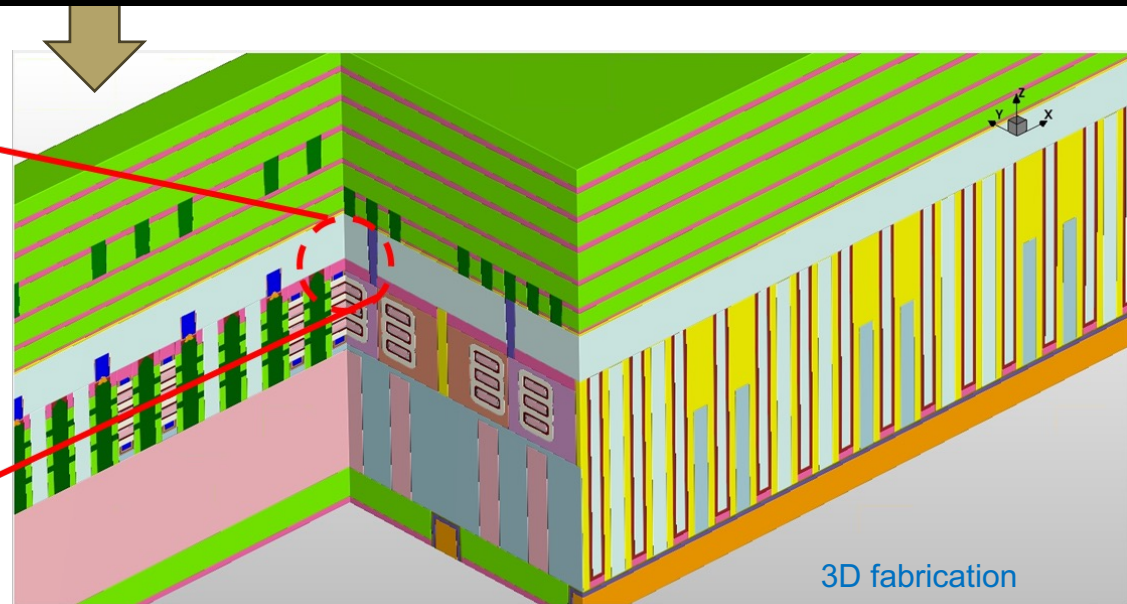
Process Explorer Modeling – D Flip-Flop



2D layout



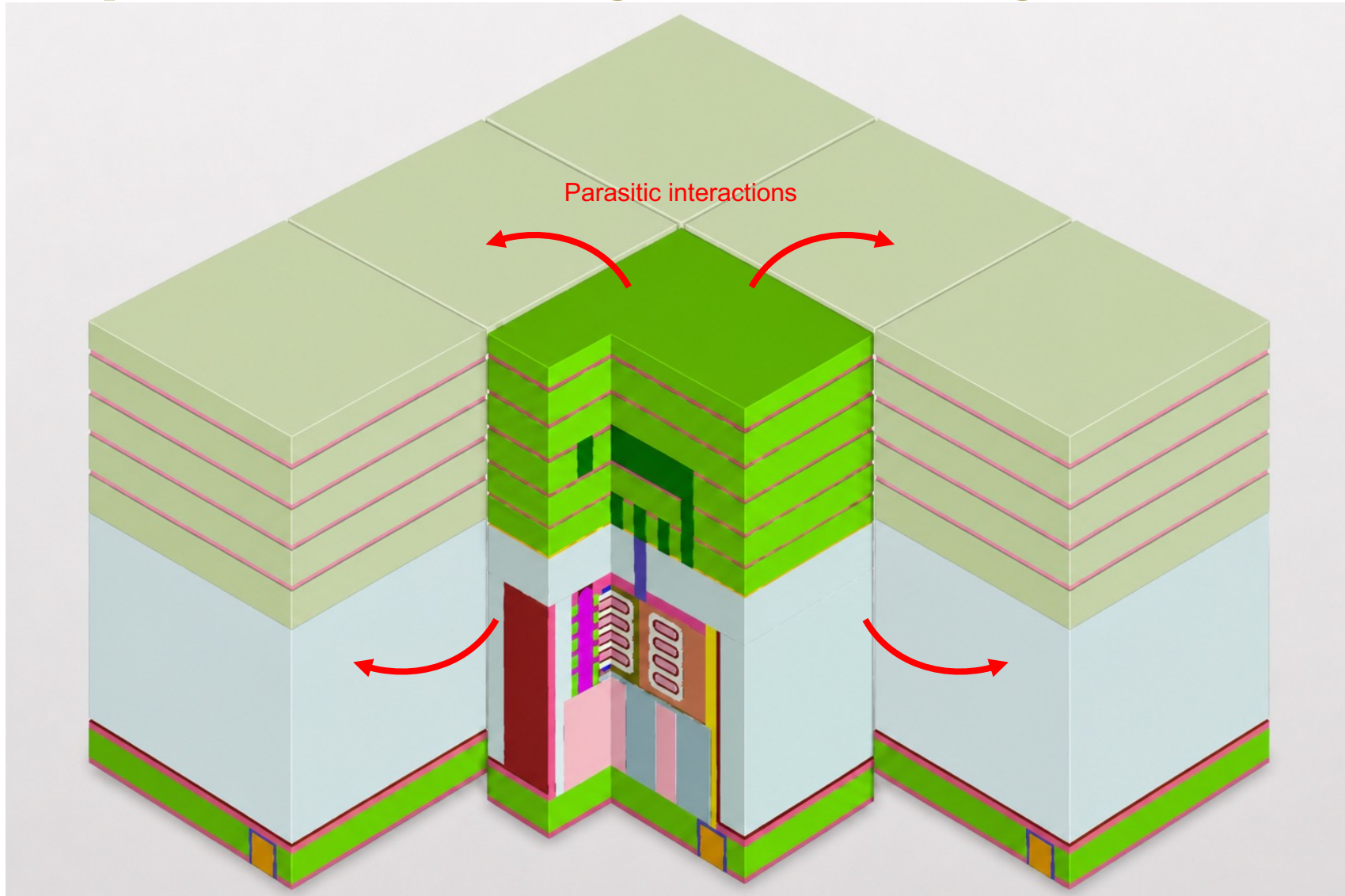
Complex gate metal structure



3D fabrication

Process Explorer flow also applicable to much more complex cell structures and processes

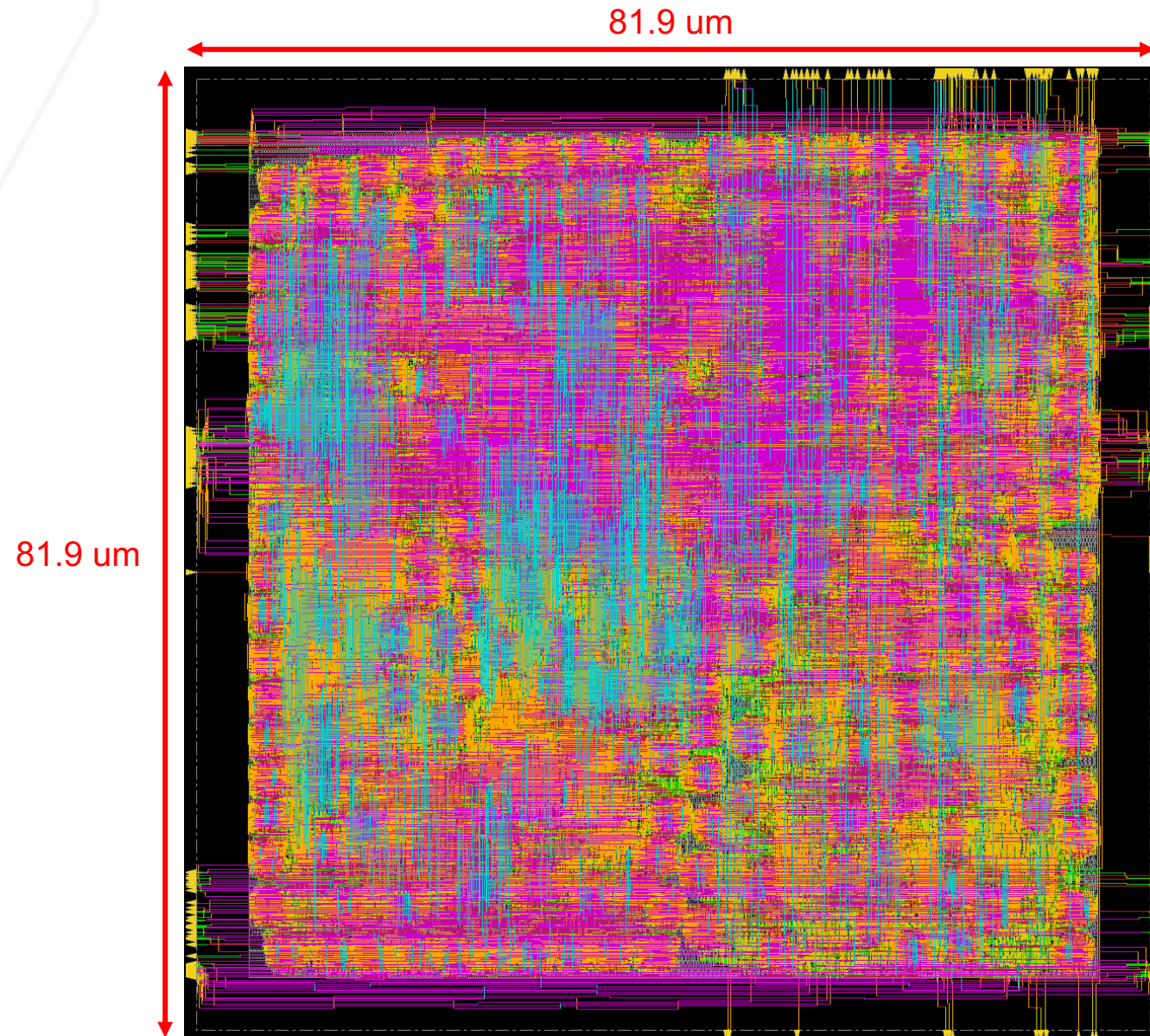
Process Explorer Modeling – Mirroring Effect



Process Explorer flow also simulates neighboring 3D cells across adjacent directions.

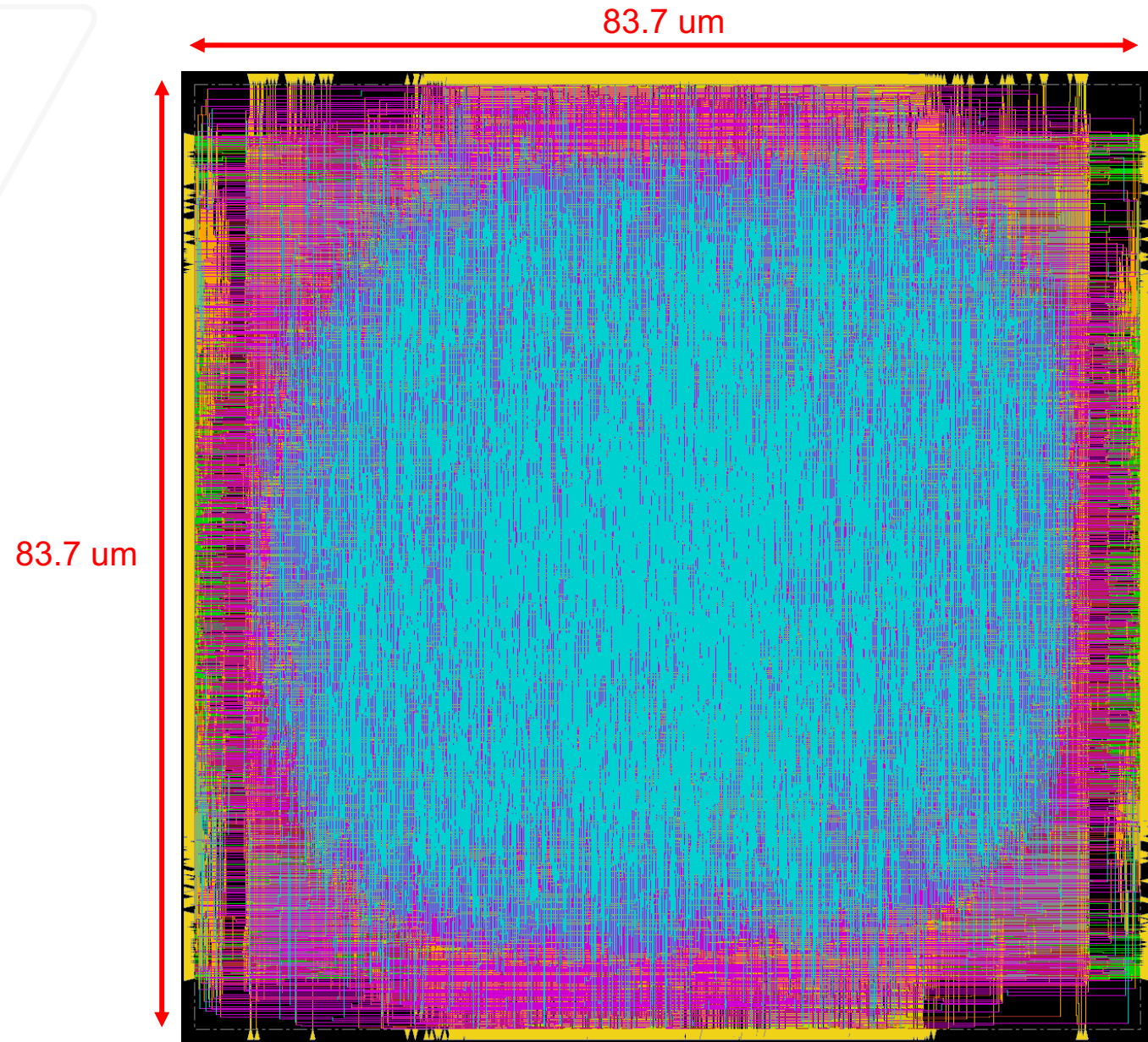
Multi-Width/VT DTCO & Benchmarking

AES 128 Bit Encryption P&R



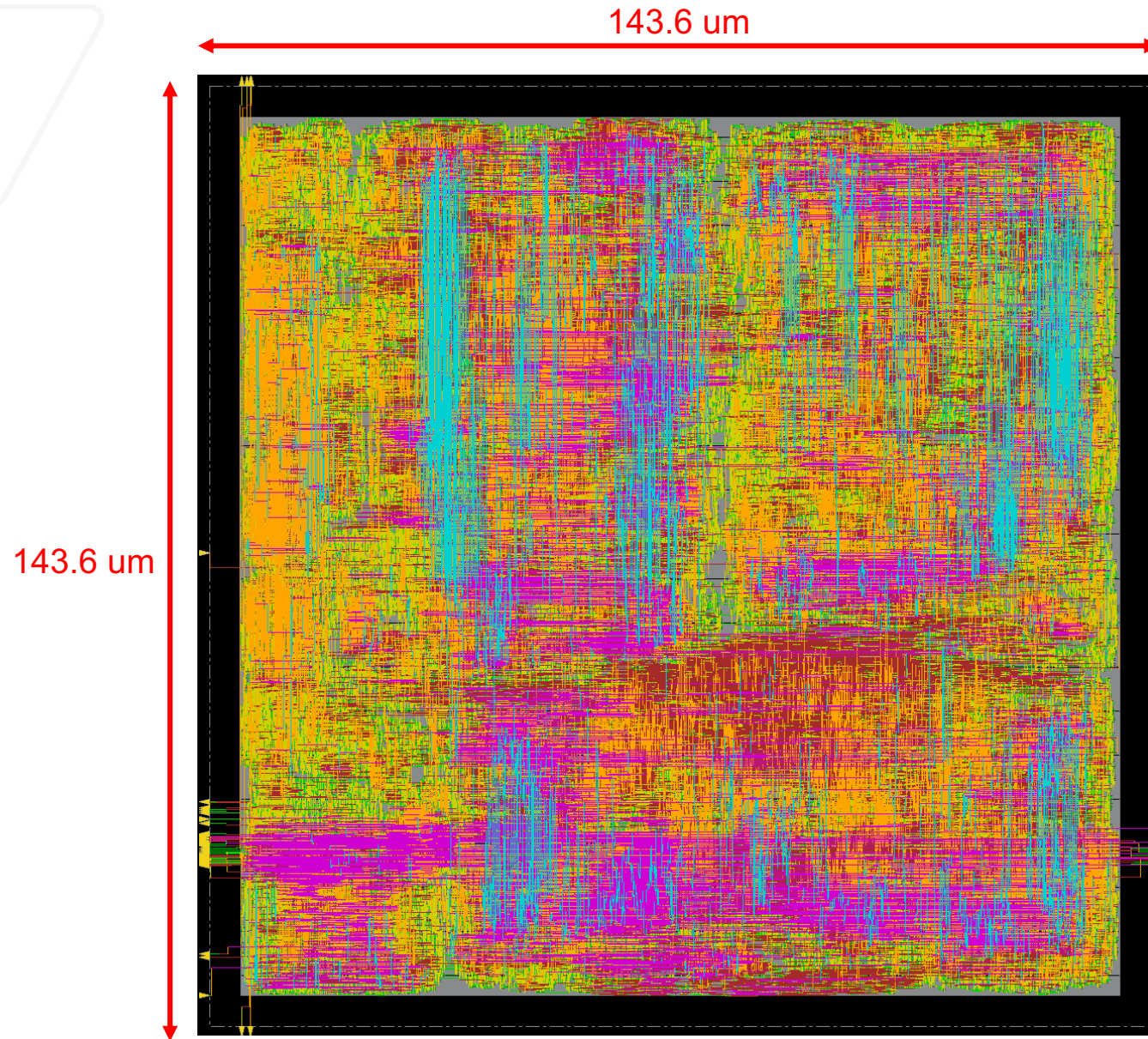
Final 2nm design (M7, 6.67 GHz clock freq.)
10 W, 10 VT flavors, 202k cells

LDPC 802.3an Decoder P&R



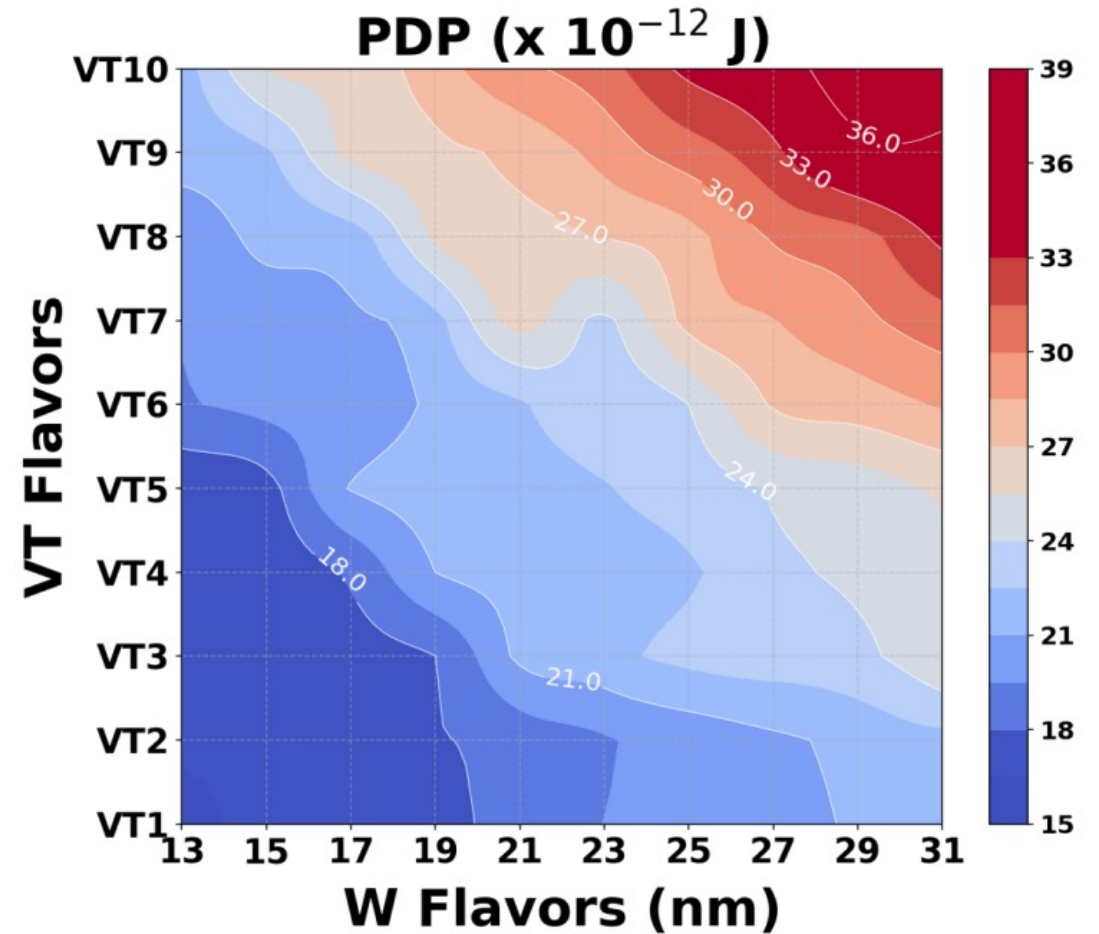
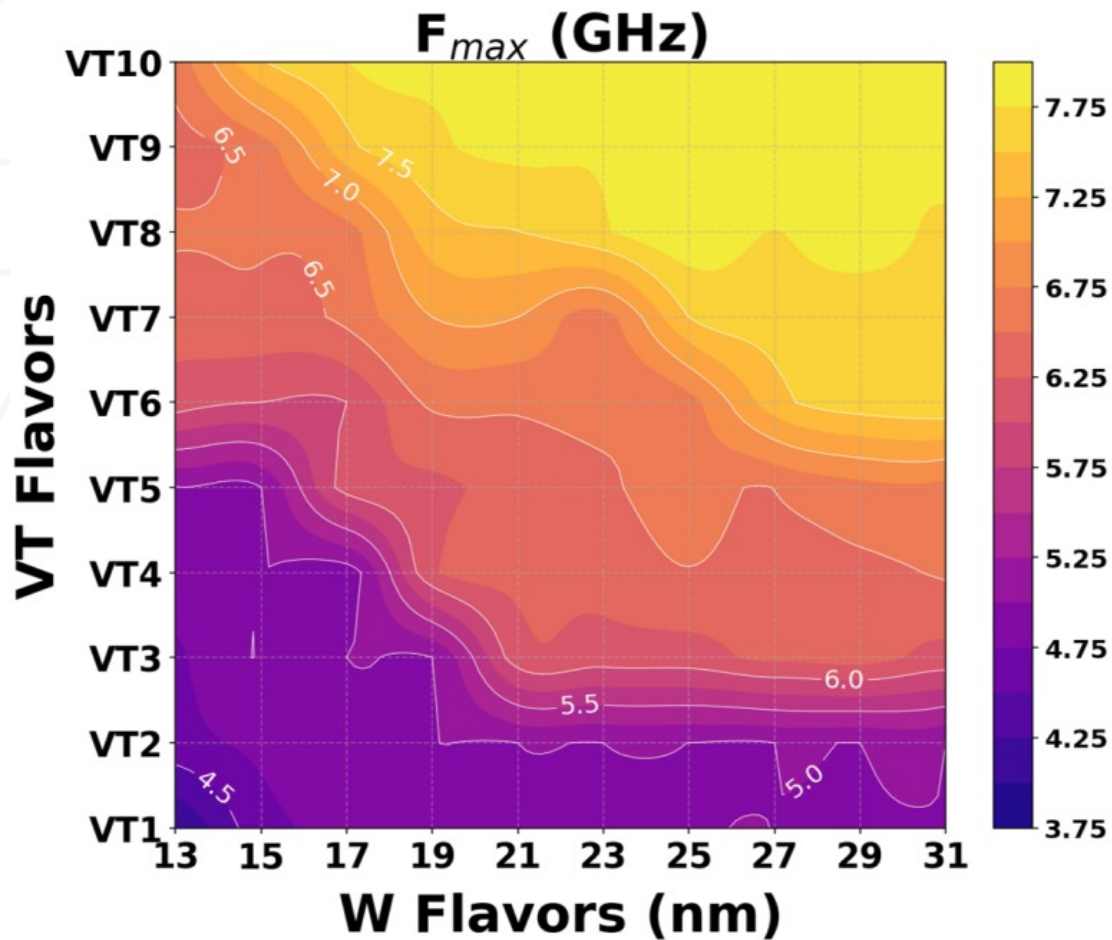
Final 2nm design (M7, 3.33 GHz clock freq.)
10 W, 10 VT flavors, 468k cells

JPEG Encoder P&R



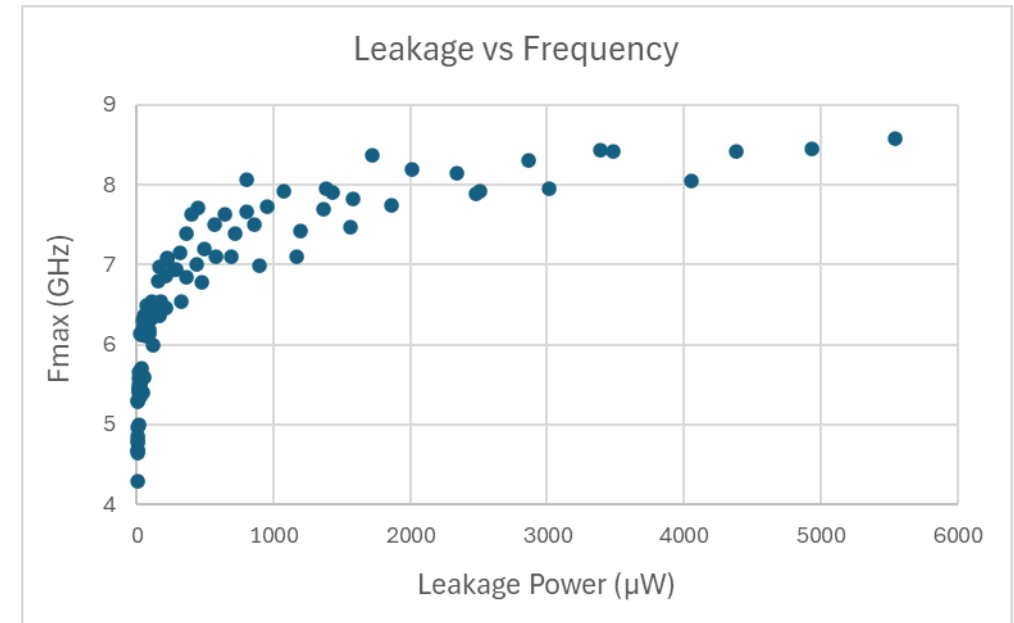
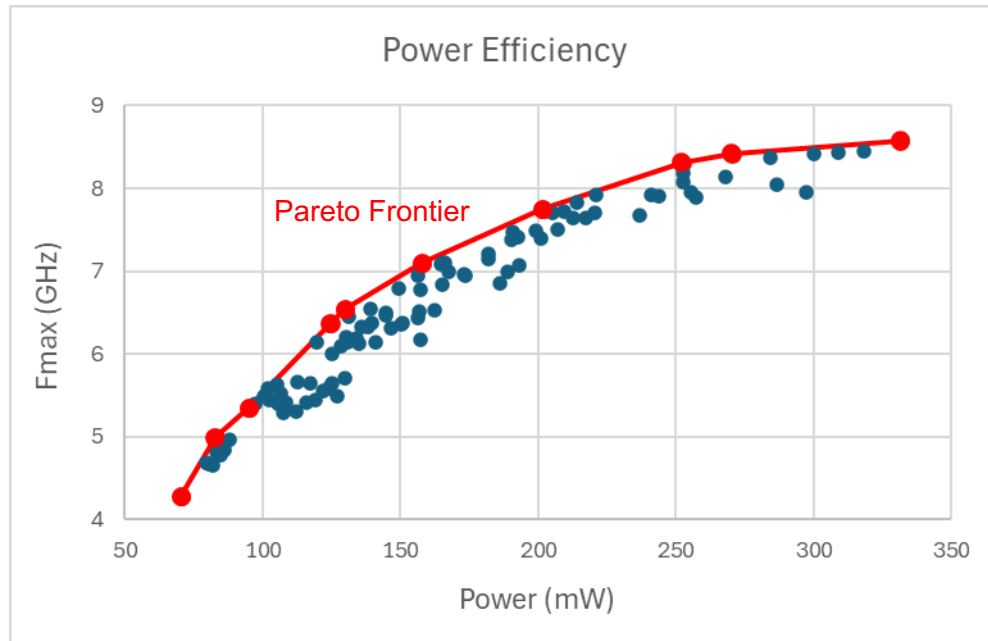
Final 2nm design (M7, 5.00 GHz clock freq.)
10 W, 10 VT flavors, 894k cells

PPA – AES Designs with Single W & VT Corners



Max Frequency & PDP contour plots vs W & VT corners

PPA – AES Designs with Single W & VT Corners

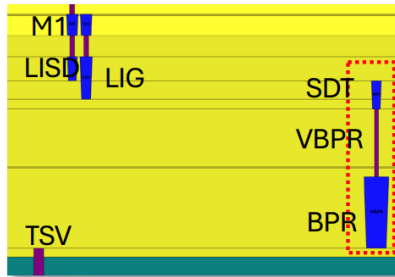


* The Pareto frontier represents the optimal (lowest-power for each frequency) corners at max-performance.

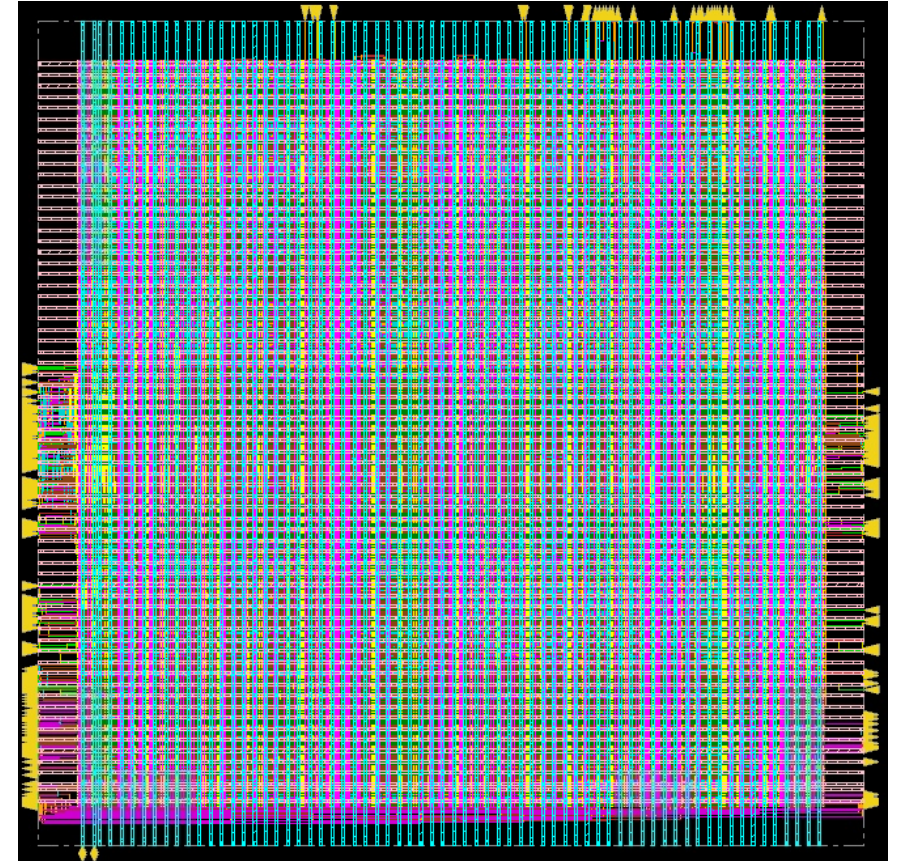
10 Best libraries: 2 nanosheet widths (13, 31 nm) & 5 VT flavors (HVT, SVT, LVT, ULVT, ELVT)

Backside PDN Implementation

NanoTSV for BM1-to-BPR

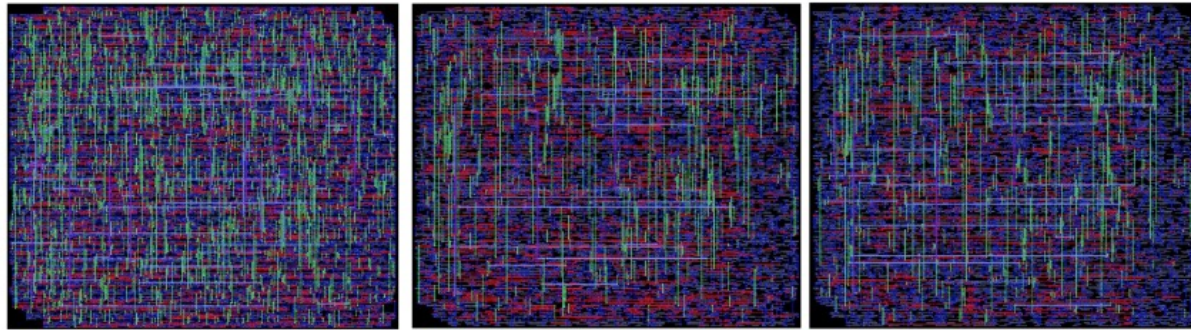


Backside metal layers for PDN
(VDD & VSS stripes)



P&R including BSPDN routing

Cell Layout Automation (SAT-Based Solver Compatibility)



Multiplexor32

Multiplexor16

Adder64

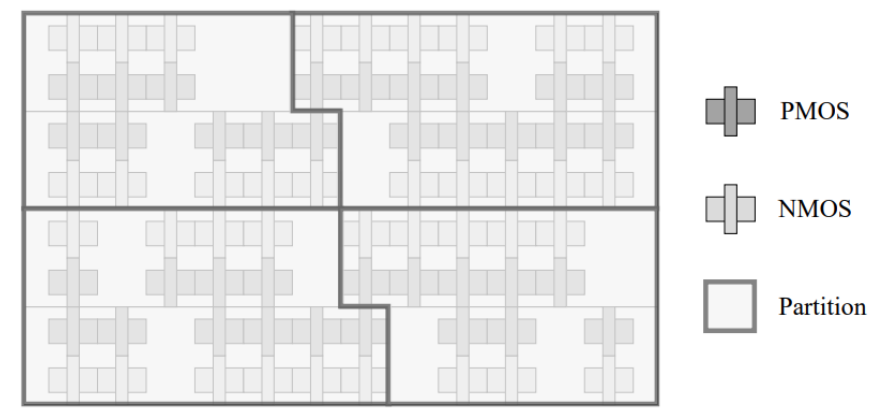


Fig. 3. Four partitions of the transistor placement.

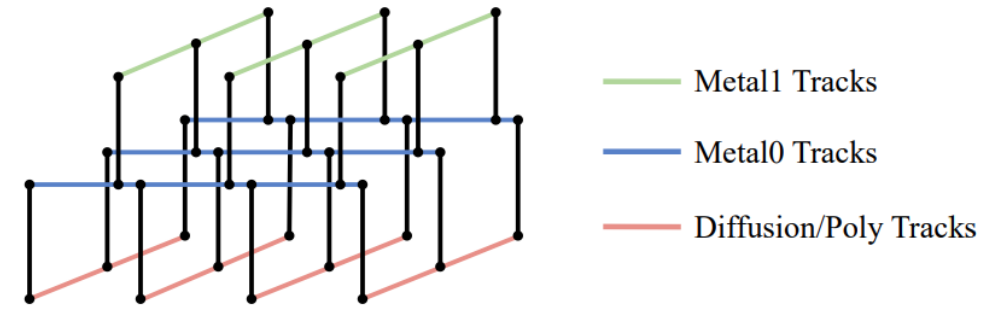


Fig. 5. A grid-based routing graph.

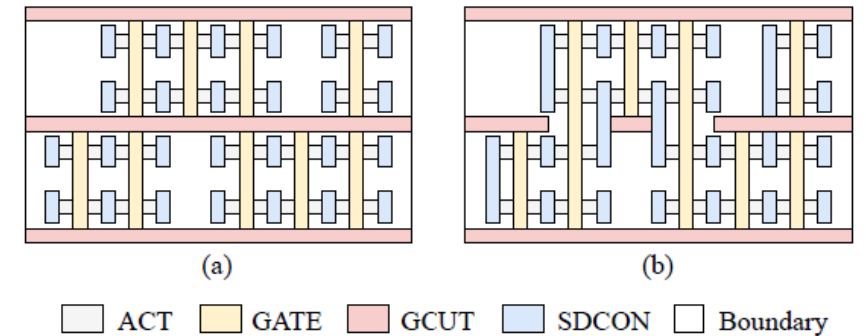
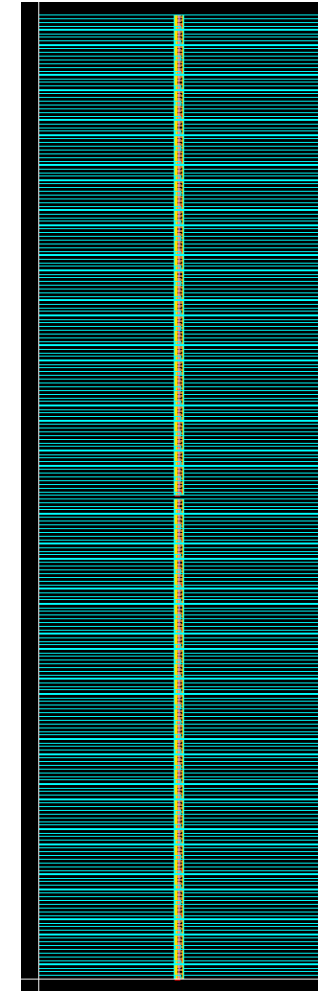
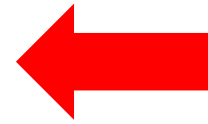
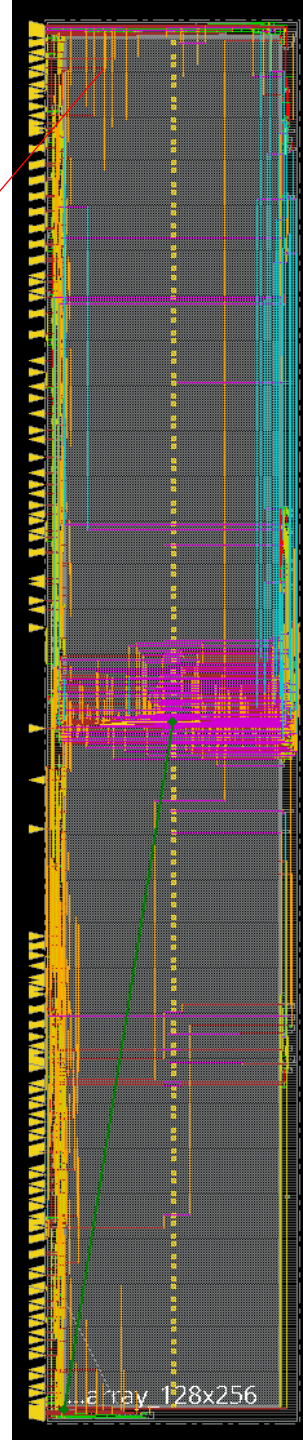


Figure 4: (a) Without direct connections across rows. (b) With direct connections across rows.

SRAM Macro P&R

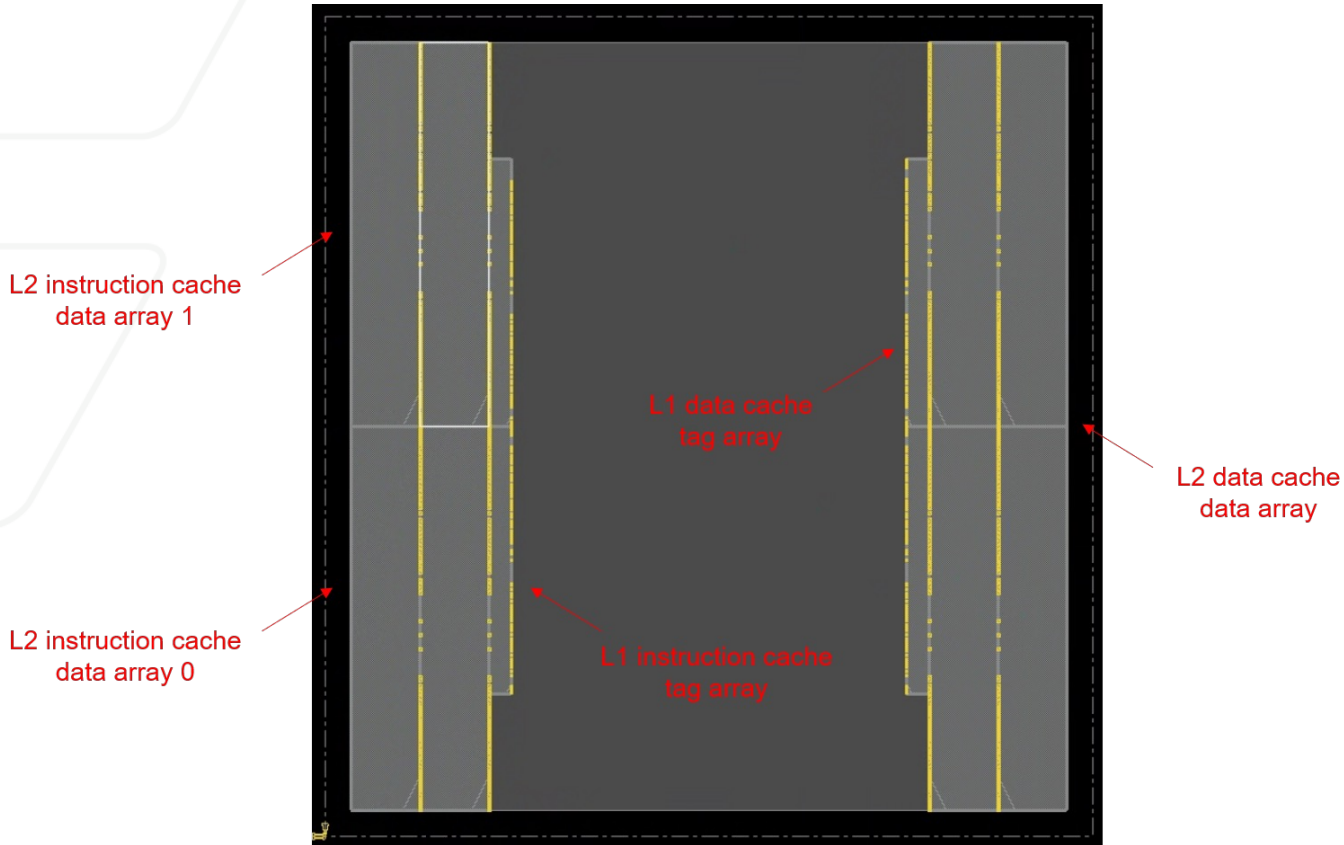


512x64 bit SRAM macro

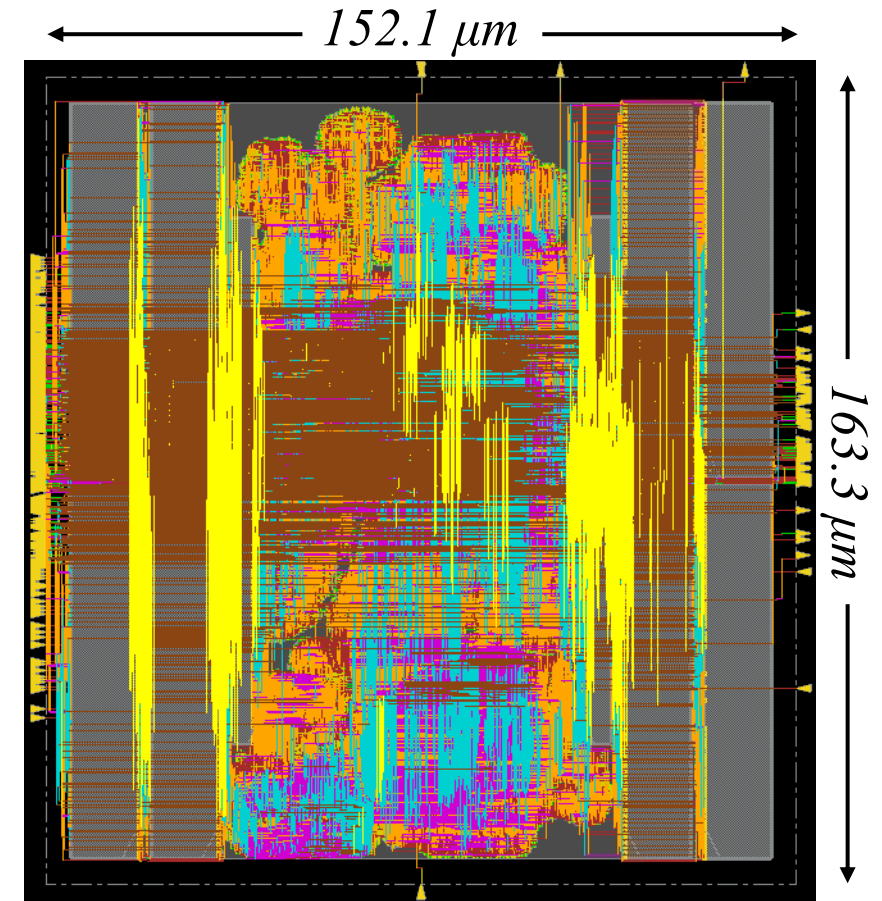


SRAM array layout design

RocketCore P&R with Generated SRAM Macros



Floorplan

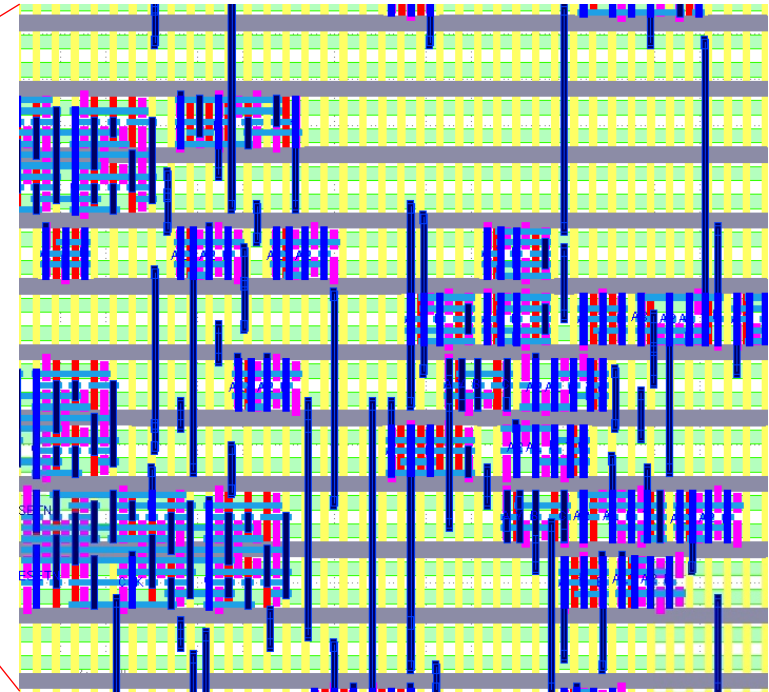


P&R GDS Layout

OpenROAD P&R Implementation with GT2N



GDS layout with complete power delivery network



Standard cell placements & M0-M1 routing

Summary & Future Plans

- 2nm nanosheet device models (provided by Synopsys)
- 3D virtually fabrication of interconnects & standard cells
- Fully-native DRC & LVS decks (advised by Synopsys & Samsung Texas Austin)
- Multi-Width & Multi-Vth offerings
- Layout-automation-solver compatible (verified by UCSD, UT Austin)
- BSPDN implementation completed
- Future plans would be
 - Incorporating a 2nm node SRAM compiler
 - SRAM DTCO with BEOL transistor implementation
 - Process corner variations (FF, SS)